



BIOINFORMATICS & DATA SCIENCE | SESSION 1 OF 2

From biological data to algorithms

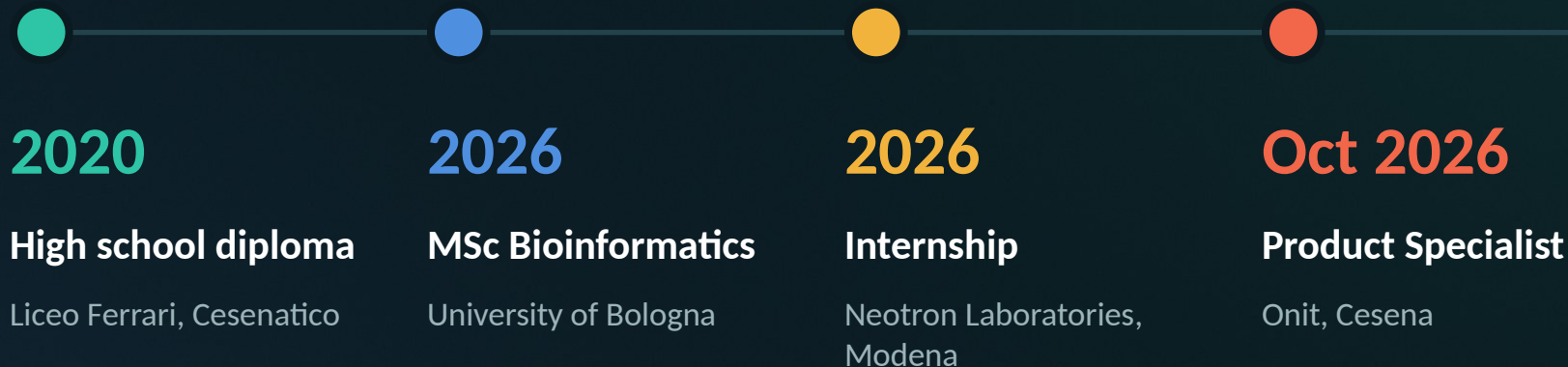
How a computer reads, compares and searches DNA

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Six years ago, I was sitting where you are



Three hours, five quizzes, two challenges

8:00 Kick-off quiz

8:15 DNA is data

8:45 Comparing sequences

9:15 Break

9:30 Challenge A: alignment by hand

10:00 Open data and code

10:10 Challenge B: sequence hunt

10:30 Live coding: from DNA to protein

10:42 Take it home, final ranking

No phones. One laptop per team.

Niente telefoni: un PC per squadra, cartellino digitale



1

laptop per team

30 s

to decide together

1 pt

per correct answer

Open the Team answer card on your laptop. Tap a letter, turn the screen to the stage.

Answer live on Wooclap

Un PC per squadra, niente telefoni

01

Open

wooclap.com on your team PC

02

Join

Enter the event code

03

Vote

One answer per team, 30 seconds

EVENT CODE

XXXXXXXX

app.wooclap.com

Where does a bioinformatician spend most of the day?

A

In the wet lab, pipetting

B

At the computer, with code and data

C

On the ward, with patients

D

At the microscope

A blue square with rounded corners containing a large, bold, black letter 'B'.

At the computer

Data comes from the lab. The job is turning it into answers, with computing, statistics and maths.

SECTION

01

DNA is data

Il DNA è informazione



A text written with four letters

Un testo scritto con quattro lettere



Start of the human beta-globin gene (HBB)

To a computer, DNA is a **string**. It can be read, searched and compared.

Four letters, two bits

Quattro lettere, due bit

A	00
C	01
G	10
T	11

HUMAN GENOME

3.1 billion bases

$\times 2 \text{ bits} \div 8 = ?$

How many bytes? Find out in the quiz.

How big is one copy of the human genome?

A

7.7 KB

B

77 GB

C

775 MB

D

7.7 TB

C

775 MB. About one CD.

The genome itself is small. Raw sequencer data is not: every base is read dozens of times, so files reach hundreds of gigabytes.

Reading DNA keeps getting cheaper

Leggere il DNA costa sempre meno

13 years

The first human genome, 1990 to 2003

1 day

A genome today, for under \$1,000

Generating data is easy. Understanding it is the job.

From your biotech chapter to the computer

Dal capitolo di biotecnologie al computer

01

PCR

Copies the DNA region millions of times

02

Sequencing

Reads millions of short fragments

03

FASTQ files

Text: letters plus a quality score

04

Alignment

Puts each read in its place

05

Variants

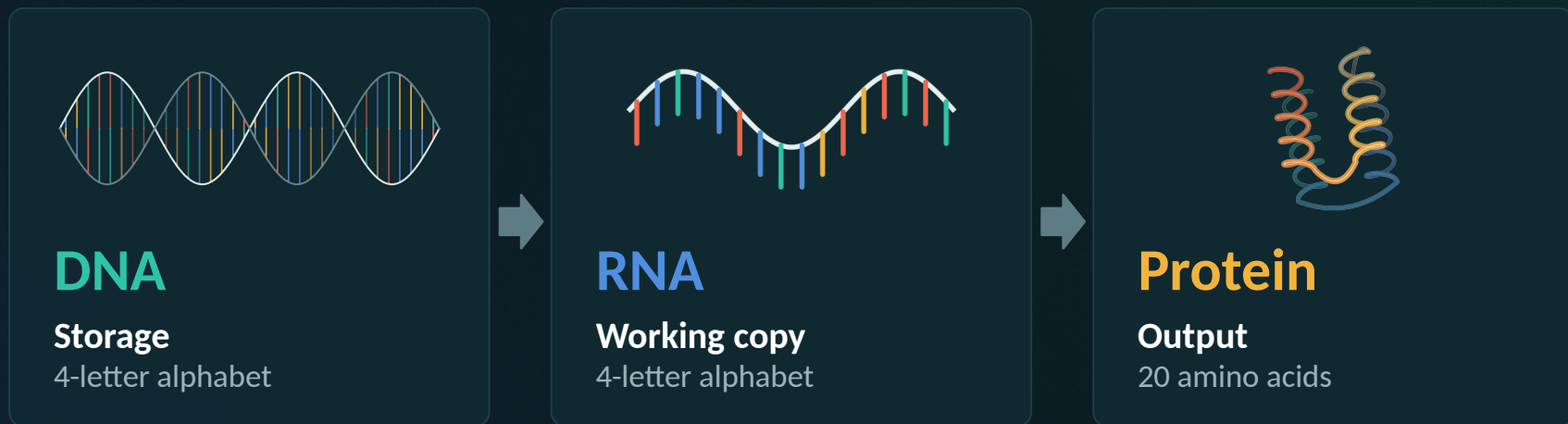
What differs from the reference?

Lab first two steps. **Bioinformatics** everything after.

Your textbook stops at the sequencer. The data starts there.

The central dogma, through a programmer's eyes

Il dogma centrale visto da un informatico



The genetic code is a **lookup table**: 3 letters in, 1 amino acid out.

A codon is 3 bases long. How many different codons exist?

A

64

B

12

C

20

D

81

A

$$64 = 4 \times 4 \times 4$$

But there are only 20 amino acids. Several codons code for the same one, and 3 of them mean stop.

SECTION

02

Comparing sequences

Confrontare sequenze



Why compare sequences?

Perché confrontare due sequenze



Identify

Whose sequence is this?



Diagnose

Is there a mutation?



Reconstruct

Who is related to whom?



Predict

What does this protein do?

The FASTA format

Il formato standard per le sequenze

```
>HBB_HUMAN hemoglobin subunit beta  
MVHLTPEEKSAVTALWGKVNDEVGGEALG  
RLLVVYPWTQRFFESFGDLSTPDAVMGNPK
```

Header starts with >, holds name and description

Sequence everything below, one letter per residue

Alignment: line them up, then score

Allineare: mettere in colonna e dare un punteggio

G	A	A	T	T	C
G	-	A	T	T	A
+1	-2	+1	+1	+1	-1

Score +1

+1

Match same letter

-1

Mismatch different letter

-2

Gap missing letter

One grid instead of a million guesses

Una griglia al posto di mille tentativi

	-	G	A	C
-	0	-2	-4	-6
G	-2	1	-1	-3
A	-4	-1	2	0
T	-6	-3	0	1
C	-8	-5	-2	1

Each cell = the maximum of



diagonal

+1 or -1



from above

-2



from the left

-2

Traceback: walk back from the end

Poi si risale dall'ultima cella

	-	G	A	C
-	0	-2	-4	-6
G	-2	1	-1	-3
A	-4	-1	2	0
T	-6	-3	0	1
C	-8	-5	-2	1

GATC
GA-C

+1

Always the best alignment, without trying them all. This is **dynamic programming**.

Exact is not the same as fast

Esatto non vuol dire veloce

20

CELLS

the two sequences above

1 million

CELLS

two genes of 1,000 bases

10^{15}

CELLS

one gene against a global
database

The shortcut

La scorciatoia

01

Words

Split the query into short words

02

Seeds

Find exact matches in the database

03

Extend

Grow the alignment around seeds

04

E-value

How likely is this by chance?

Exact and slow, or fast and almost always right.

Why is BLAST so much faster?

A

It runs on faster computers

B

It starts from short exact matches, not the full grid

C

It only checks the first 10 letters

D

It ignores gaps

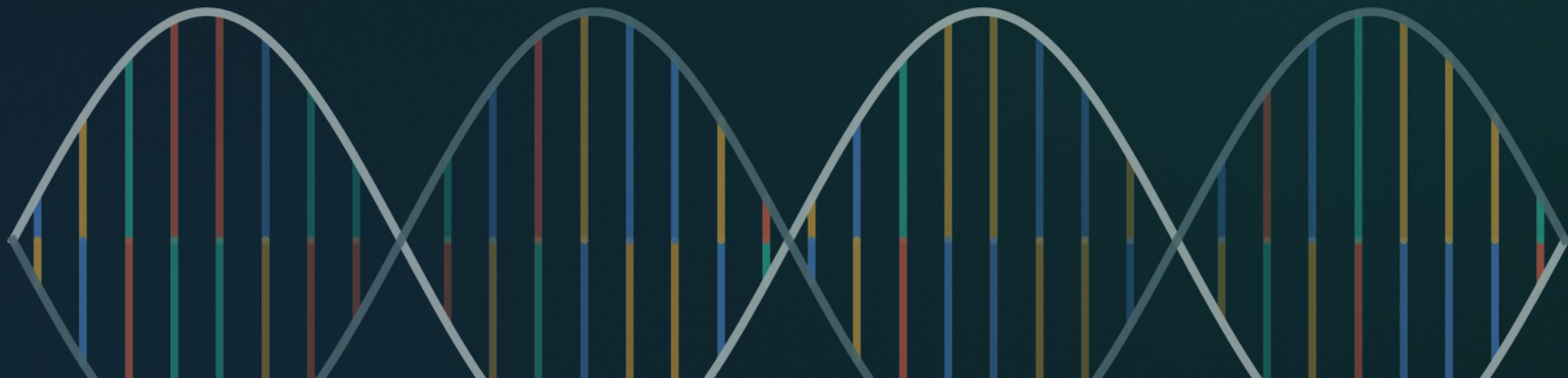
A blue square with rounded corners containing a large, bold, black letter 'B'.

It skips most of the grid

It only computes around the seeds. It may miss weak similarities, but it answers in seconds on huge databases.

Break

15 minutes. Then it's your turn.



Alignment by hand

Sfida A, scheda A. Allineamento su carta

01

Set up

First row and column: 0, -2, -4...

02

Fill

Max of the three neighbours

03

Trace back

From the last cell back to 0

Warm-up GATC vs GAC

Challenge GAATTC vs GATTA

Two right answers

Due soluzioni con lo stesso punteggio

	-	G	A	T	T	A
-	0	-2	-4	-6	-8	-10
G	-2	1	-1	-3	-5	-7
A	-4	-1	2	0	-2	-4
A	-6	-3	0	1	-1	-1
T	-8	-5	-2	1	2	0
T	-10	-7	-4	-1	2	1
C	-12	-9	-6	-3	0	1

GAATTC
GA-TTA

+1

highlighted path

GAATTC
G-ATTA

+1

second path

SECTION

03

Open data and code

Dati pubblici e codice



Free for anyone, anywhere

Archivi pubblici e gratuiti



NCBI

DNA and RNA sequences.
Home of BLAST.

ncbi.nlm.nih.gov ↗



UniProt

Proteins, their function
and linked diseases.

uniprot.org ↗



PDB

3D structures. Over
200,000 of them.

rcsb.org ↗

Python, R and SQL

I linguaggi del mestiere



pipelines, automation, machine learning

```
seq = "ATGCGC"  
gc = seq.count("G") + seq.count("C")  
print(gc / len(seq))
```



statistics and plots

```
x <- c(1200, 850, 3100)  
mean(x)  
hist(x)
```

GC content the share of G and C in a sequence

SQL the language of databases, the one from your computer science class

What does the Python code print for "ATGCGC"?

A

0.33

B

0.50

C

0.67

D

1.00

**0.67**

Two Gs and two Cs: 4 out of 6. The code does what you would do by hand, on millions of letters per second.

Sequence hunt

Sfida B, scheda B. Un PC per squadra

01

Copy

One sequence from
sheet B

02

BLAST

blast.ncbi.nlm.nih.gov,
Protein BLAST

03

Identify

Protein and organism

04

UniProt

What does it do?

Final question Sequence 4 is almost identical to 3. What changes? What disease does it cause?

[Open BLAST ↗](#)

[Open UniProt ↗](#)

Three proteins and one mutation

Tre proteine e una mutazione

- | | | |
|---|-------------------------|-----------------------------------|
| 1 | Human insulin | controls blood sugar |
| 2 | GFP, jellyfish | glows green, used as a lab marker |
| 3 | Human hemoglobin (beta) | carries oxygen |
| 4 | Hemoglobin, E becomes V | sickle cell disease |

DNA is just a string

Il DNA per Python è una stringa

PYTHON

```
dna = "ATGGTGCACCTGACTCCTGAGGAGAAG"

print(len(dna))
print(dna.count("G"))

def gc_content(seq):
    gc = seq.count("G") + seq.count("C")
    return gc / len(seq)

print(round(gc_content(dna), 2))
```

OUTPUT

```
27
9
0.56
```

The same HBB gene from the start of today. Two built-in functions and one of our own.

The genetic code is a dictionary

Il codice genetico come tabella di conversione

PYTHON

```
bases = "TCAG"
amino = ("FFLLSSSSYY**CC*W"
        "LLLLPPPPHHQRRRR"
        "IIIMTTTTNNKKSSRR"
        "VVVVAAAADDEEGGGG")
codons = [a + b + c for a in bases for b in bases for c in bases]
code = dict(zip(codons, amino))
print(len(code))

def translate(seq):
    protein = ""
    for i in range(0, len(seq) - 2, 3):
        protein += code[seq[i:i + 3]]
    return protein

print(translate(dna))
```

OUTPUT

64
MVHLTPEEK

One letter, one disease

Una lettera, una malattia

PYTHON

```
sickle = dna[:19] + "T" + dna[20:]  
  
print(dna)  
print(sickle)  
print(translate(dna))  
print(translate(sickle))
```

OUTPUT

```
ATGGTGCACCTGACTCCTGAGGAGAAG  
ATGGTGCACCTGACTCCTGTGGAGAAG  
MVHLTPEEK  
MVHLTPVEK
```

DNA

GAG → **GTG**

AMINO ACID

E → **V**

PROTEIN

healthy → **sickle cell**

All the code and quizzes, online

Tutto il codice e i quiz, da rifare a casa



Code

every live demo, ready for Google Colab

Quiz

all the questions, with the explanation

Links

BLAST, UniProt, AlphaFold, Rosalind, Unibo

Feedback

anonymous, 2 min: link from your teachers

[Toolkit page: scan the QR ↗](#)

Three things to remember

1

DNA is text. A few lines of code can read it.

2

Exact or fast: you usually have to choose.

3

The world's biological data is open. You just used it.

Next session How a computer learns